



TECHNICAL ARSENAL

LANGUAGES

Python, TypeScript, JavaScript (ES6+), SQL, Bash

BACKEND & SYSTEMS

FastAPI, Node.js, Express, REST APIs, Webhook Reliability (HMAC-SHA256, Idempotency, DLQ, Jitter)

DISTRIBUTED & AI

DAG Engines, Write-Ahead Logs (CRC32, Merkle Proofs), Prompt Injection Guards, Token Bucket, Circuit Breakers

DATABASES & OPS

PostgreSQL, SQLite, Docker, GitHub Actions (CI/CD), Zod, Playwright, pytest, Vitest

EMPIRICAL METRICS

HOOKSENTINEL

6,661 req/s · 0.58ms p99

PROMPTSHIELD

1.64M scans/s · 0.4µs

RESILIENTDAG

179k graphs/s · 4.2µs

CRASHPROOFWAL

847k writes/s · 15ms rec

ENGINEERING FOCUS

- Cryptographic webhook safety
- Crash-resilient write pipelines
- Topological DAG orchestration
- Enterprise LLM security proxy
- Zero layout shift frontend

EDUCATION

RMIT University Vietnam

Bachelor of Business · Systems Focus
2021 – 2025 · Ho Chi Minh City
Applied software systems, distributed architectures & API engineering.

VERIFIED OPEN SOURCE & FLAGSHIP SYSTEMS

python-docx-ng — Quote-Safe XPath Bugfix

MERGED UPSTREAM

Aug 2026

Upstream Contributor · PR #131 · Python XPath unittest CI

- **Upstream Core Fix:** Resolved style-lookup defect where single quotes in Word style names broke XPath query evaluation across document trees; replaced interpolated raw XPath strings with bound parameter lookups.
- **Engineered Fix & Test Matrix:** Authored regression test suite; independently reviewed and merged into upstream release.

webhook-gateway — Ingestion & Dead-Letter Broker

Sep 2026

Author · 78 Tests · Live Demo ↗ · FastAPI Python SQLite WAL Docker

- **Empirical Ingest Benchmark:** Engineered multi-provider HMAC verification (Stripe, GitHub, Shopify) achieving 6,661 req/s with sub-millisecond p99 latency (0.58ms).
- **Replay Attack Shield:** Intercepted 100% of concurrent replay bursts (11,904 checks/s) via deterministic idempotency locks (`provider:sha256(body)`); isolated poisoned deliveries to DLQ.

streamline-engine — Resilient DAG Data Pipeline

Sep 2026

Author · 10 Tests · Live Demo ↗ · TypeScript Node.js Kahn DAG Zod

- **Topological DAG Solver:** Implemented Kahn's algorithm resolving execution tiers at 179,421 graphs/s (4.2µs p50) with 100% circular deadlock interception (115k checks/s).
- **Zero-Loss Partial Resumption:** Enforced runtime Zod schema boundaries; built disk checkpointing saving 100% of upstream compute on mid-pipeline crashes.

ai-guard-gateway — LLM Security Proxy & Fallback Router

Sep 2026

Author · 7 Tests · Live Demo ↗ · TypeScript OWASP LLM01 Circuit Breaker

- **OWASP LLM01 Security Guard:** Built heuristic scanner blocking DAN jailbreaks and system overrides at 1.64M scans/s (0.4µs latency); added token-bucket rate limiter.
- **Semantic Cache & Fallback:** Engineered exact-match SHA-256 caching delivering 690k reads/s (<1µs hits, 100% token savings); automated circuit breaker failover.

wal-kv-ledger — Crash-Resilient Write-Ahead Log Store

Sep 2026

Author · 5 Tests · Live Demo ↗ · Python CRC32 Merkle Tree

- **Binary Append-Only Durability:** Engineered binary storage engine with hardware CRC32 checksums processing 847,449 writes/s (72.4 MB/s) and in-memory index serving 2.7M reads/s.
- **Surgical Crash Recovery:** Built automatic corrupted-tail truncation recovering from `SIGKILL` in 15.03 ms with 0 bytes committed data lost; integrated SHA-256 Merkle tree state proofs.

ENGINEERING EXPERIENCE

Independent Software Engineering Practice

Aug 2023 – Present

Senior Software Engineer · Ho Chi Minh City, Vietnam · Node.js Express PostgreSQL

- **60-Second Lead Capture Engine:** Architected durable lead ingestion service (Node.js/Express, rate limiting, Telegram webhook alerts, write-ahead failover logging) delivering sub-minute response times and failover logging.
- **Automotive Workforce Attendance System:** Built operations web application for an automotive manufacturing plant in Vietnam, translating complex shift scheduling and punch-in rules into validated data models.
- **ERP Data Pipeline & Reporting:** Engineered automated ETL reconciliation scripts extracting transaction records from Odoo ERP and spreadsheets into audited multi-format reports (Excel, HTML, PDF).

SkillArena — Interactive Simulation Platform

2026 – Present

Full-Stack Lead Builder · Next.js React TypeScript Supabase Zod

- **Full-Stack Architecture:** Engineered interactive simulation platform with provider abstraction, strict Zod validation, and Playwright end-to-end test suites.